Matrix functions and stability

# Degree coverage with seven matrix multiplications

**Difficulty:** challenging **Importance:** interesting to the community

**Status: Open**

**Rating rationale:** Challenging because numerical evidence for coefficient-space coverage needs exact algebraic justification; community impact comes from multiplication budgets for practical matrix-function evaluation.

## Problem statement

Start with the scalar polynomials $1, x \in \mathbb{C}[x]$. Allow arbitrarily many complex linear combinations of already computed polynomials and at most seven multiplications of two already computed polynomials. Let $\mathcal{P}_7 \subseteq \mathbb{C}[x]_{\leq 128}$ be the set of possible outputs.

Identify a polynomial with its coefficient vector in $\mathbb{C}^{129}$ and let $\overline{\mathcal{P}_7}^Z$ be its Zariski closure: the common zero set of all polynomial equations in the coefficients that vanish throughout $\mathcal{P}_7$. Is

$$\max\{d \in \{0, \dots, 128\} : \mathbb{C}[x]_{\leq d} \subseteq \overline{\mathcal{P}_7}^Z\} = 42?$$

The target concerns closure and the complex coefficient field. Exact representability of every polynomial is a stronger assertion.

## Why it matters

For a dense input matrix, matrix products dominate the cost of polynomial evaluation. The conjecture determines how much polynomial degree a budget of seven products can cover, providing a theoretical benchmark for matrix-function evaluation schemes.

### References


1. E. Jarlebring and G. Lorentzon, *The polynomial set associated with a fixed number of matrix-matrix multiplications*, arXiv:2504.01500v3 (13 August 2025), §2.2 (closure convention), §5.4, Conjecture 13. Primary text.
2. J. Sastre, J. Ibáñez, J. M. Alonso, and E. Defez, *Beyond Paterson–Stockmeyer: Advancing matrix polynomial computation*, WSEAS Transactions on Mathematics 24 (2025), 684–693, §4 (a five-product construction). Primary paper, DOI.
3. J. M. Alonso, J. Sastre, J. Ibáñez, and E. Defez, *A systematic framework for stable and cost-efficient matrix polynomial evaluation*, arXiv:2603.23143v1 (24 March 2026), abstract and conclusions. Primary text.


### Status check — 2026-09-08

The latest arXiv source remains v3 and labels the displayed equality a conjecture supported by computations. The later papers give constructions and stability procedures without establishing the seven-product degree-coverage claim. Searches included the exact title with *2026*, *Jarlebring Lorentzon conjecture seven multiplications 42*, and *Sastre polynomial evaluation 2026*. No resolution was located. A 3 September 2026 author talk also describes higher-cost degree coverage as conjectural. Six-product real/complex formulations have a separate source ambiguity and are deliberately not included here.

### Audit — 2026-09-10

Rechecked Conjecture 13 and Remark 14: complex closure coverage at seven products is still conjectural, with concrete matrix-exponential applications. Title and seven-product searches found no exact proof. The June 2026 degree-eight construction solves a different budget/degree problem.